

Analog Integrated Circuit Design

Cadence Tools

Lab 02

Common Source Amplifier

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1 Part 1: Sizing Chart

1.1 Design Specifications

The target specifications for the 0.18 μm CMOS technology are:

Table 1: Design specifications.

Parameter	Value
DC Gain $ A_v $	10
Supply V_{DD}	1.8 V
Current consumption I_{DQ}	80 μA
Channel length L	2 μm

This version of the generator creates a random integer. It can deal with very large integers up to a few thousand digits.

Result

80

Lower Limit 11
Upper Limit 100

Generate Clear

(a) Random number generator showing bias current = 80 μA .

This version of the generator creates a random integer. It can deal with very large integers up to a few thousand digits.

Result

10

Lower Limit 5
Upper Limit 10

Generate Clear

(b) Random number generator showing gain = 10.

Figure 1: Randomly assigned design parameters for this report.

1.2 Hand Analysis: Design Equations

1.2.1 Load Resistor R_D

To maximise output signal swing, the common-mode (CM) output is placed at $V_{DD}/2$, which means the DC voltage drop across R_D equals $V_{DD}/2$:

$$V_{R_D} = \frac{V_{DD}}{2} = \frac{1.8}{2} = 0.9 \text{ V} \quad (1)$$

Given the DC bias current $I_{DQ} = 80 \mu\text{A}$:

$$R_D = \frac{V_{R_D}}{I_{DQ}} = \frac{0.9}{80 \times 10^{-6}} = 11.25 \text{ k}\Omega \quad (2)$$

1.2.2 Required V_Q^*

The small-signal voltage gain of the resistively loaded CS amplifier (ignoring r_o for now) is

$$|A_v| \approx g_m R_D = \frac{2I_D}{V^*} R_D = \frac{2V_{R_D}}{V^*} \quad (3)$$

where $V^* = 2I_D/g_m$ is the effective overdrive (equal to V_{ov} only for a square-law device). Rearranging for the required V^* :

$$V_Q^* = \frac{2V_{R_D}}{|A_v|} = \frac{2 \times 0.9}{10} = 0.18 \text{ V} \quad (4)$$

1.2.3 Transistor Sizing: Width W

Because $I_D \propto W$ at any fixed V_{GS} (irrespective of the operating model), the sizing chart generated with $W_{\text{ref}} = 10 \mu\text{m}$ gives a reference current I_{DX} at $V^* = V_Q^*$. The required width is then found by ratio and proportion:

$$\frac{W}{10 \mu\text{m}} = \frac{I_{DQ}}{I_{DX}} \implies W = 10 \mu\text{m} \times \frac{I_{DQ}}{I_{DX}} \quad (5)$$

1.3 Simulation Results: Sizing Chart

1.3.1 V^* and V_{ov} vs. V_{GS} where $V_{th} = 0.383V$

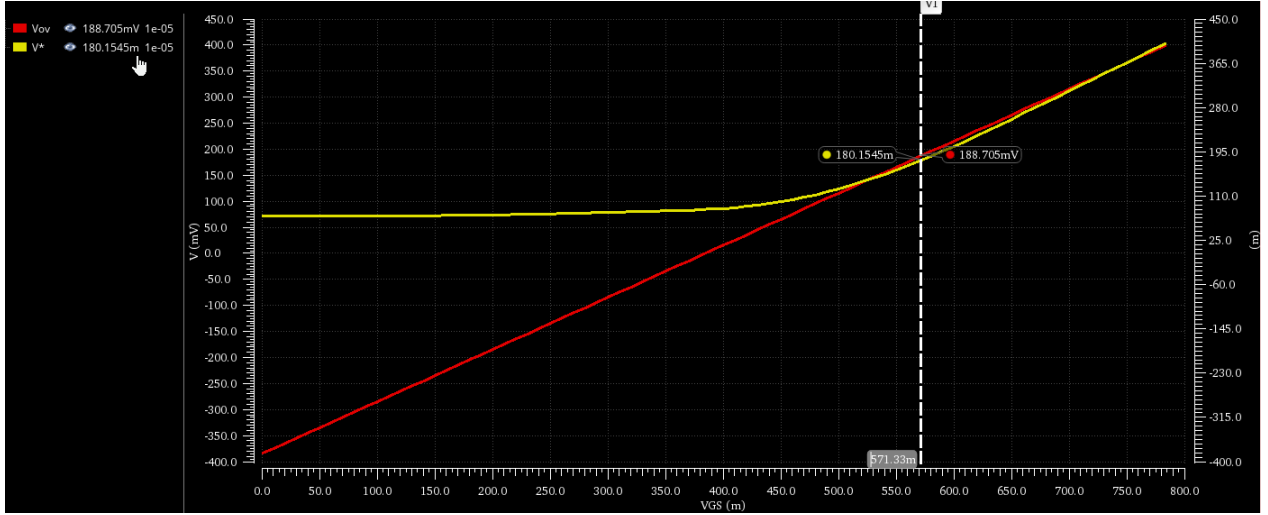


Figure 2: $V^* = 2I_D/g_m$ (Yellow) and $V_{ov} = V_{GS} - V_{TH}$ (Red) vs. V_{GS} for the NMOS with $W = 10 \mu\text{m}$, $L = 2 \mu\text{m}$.

1.3.2 I_D , g_m , g_{ds} vs. V_{GS}

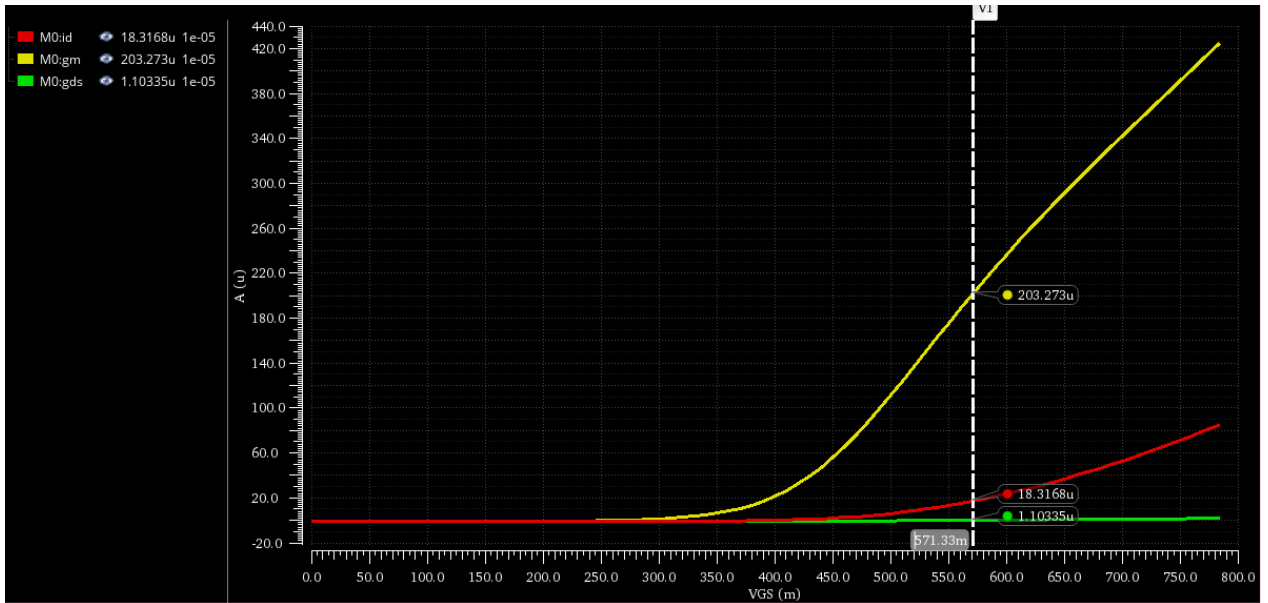


Figure 3: I_D , g_m , and g_{ds} vs. V_{GS} for the reference NMOS ($W = 10 \mu\text{m}$, $L = 2 \mu\text{m}$, $V_{DS} = V_{DD}/2$).

From the sizing chart (Figure 2), at $V_Q^* = 0.18 \text{ V}$ we read:

Table 2: Values read from the sizing chart at $V_Q^* = 0.18\text{ V}$, $W_{\text{ref}} = 10\text{ }\mu\text{m}$, $L = 2\text{ }\mu\text{m}$.

Quantity	Symbol	Value
Reference drain current	I_{DX}	$18.32\text{ }\mu\text{A}$
Reference transconductance	g_{mX}	$203.27\text{ }\mu\text{S}$
Reference output conductance	$g_{ds,X}$	$1.103\text{ }\mu\text{S}$
$V_{ov,Q}$	$V_{ov,Q}$	188.71 mV
Corresponding V_{GS}	V_{GSQ}	571.33 mV

$$W = 10\text{ }\mu\text{m} \times \frac{80\text{ }\mu\text{A}}{I_{DX}} \approx 43.67\text{ }\mu\text{m} \quad (6)$$

1.3.3 Scaled Small-Signal Parameters

Since $g_m \propto W$ and $g_{ds} \propto W$ at constant V_{ov} :

$$g_{mQ} = g_{mX} \times \frac{W}{10\text{ }\mu\text{m}} \approx 887.68\text{ }\mu\text{S} \quad (7)$$

$$g_{ds,Q} = g_{ds,X} \times \frac{W}{10\text{ }\mu\text{m}} \approx 4.82\text{ }\mu\text{S} \implies r_{oQ} = \frac{1}{g_{ds,Q}} \approx 207.6\text{ K}\Omega \quad (8)$$

1.3.4 Gain Verification

The actual gain including r_o :

$$A_v = -g_m(R_D \parallel r_o) = -887.68\text{ }\mu\text{S} \times \frac{11.25\text{ k}\Omega \times 0.2076\text{ M}\Omega}{11.25\text{ k}\Omega + 0.2076\text{ M}\Omega} = -9.5 \approx -10 \quad (9)$$

The parameters are correct!

Table 3: Final design parameters from Part 1 sizing chart ($0.18\text{ }\mu\text{m}$ CMOS, $L = 2\text{ }\mu\text{m}$, $I_{DQ} = 80\text{ }\mu\text{A}$, $|A_v| = 10$).

Parameter	Symbol	Value
Transistor Width	W	$43.67\text{ }\mu\text{m}$
Channel Length	L	$2\text{ }\mu\text{m}$
Transconductance	g_m	$887.68\text{ }\mu\text{S}$
Output Conductance	g_{ds}	$4.82\text{ }\mu\text{S}$
Output Resistance	r_o	$207.6\text{ k}\Omega$
Load Resistor	R_D	$11.25\text{ k}\Omega$
Gate-Source Bias Voltage	V_{GSQ}	571.33 mV

2 Part 2: CS Amplifier

2.1 OP and AC Analysis

2.1.1 DC Operating Point

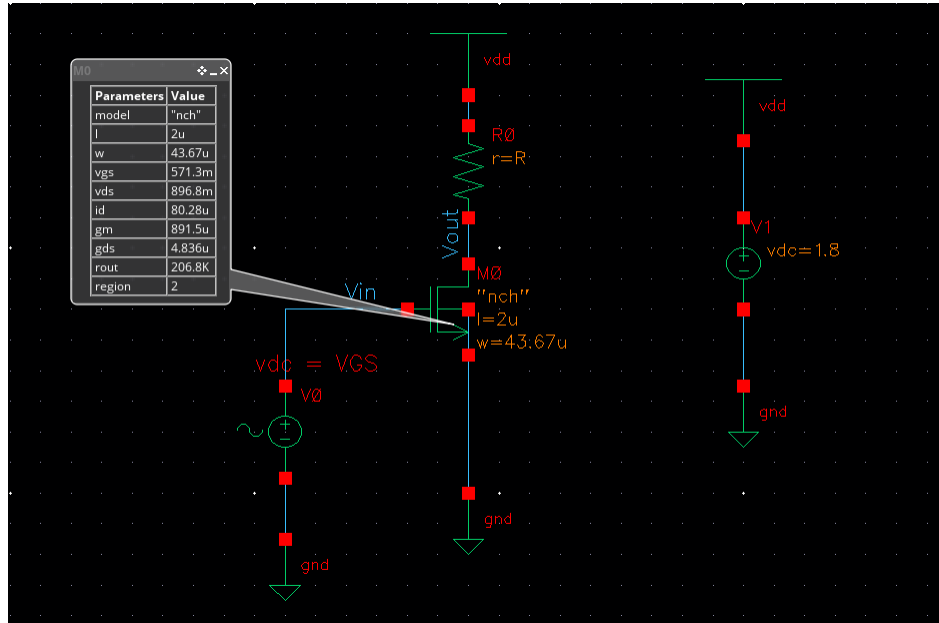


Figure 4: DC operating point annotation from Cadence Virtuoso.

Table 4: Comparison of key parameters between the sizing chart (Part 1) and DC operating point simulation (Part 2).

Parameter	Part 1 (Sizing Chart)	Part 2 (DC Simulation)	% Error
V_{GS}	571.33 mV	571.3 mV	—
I_D	80 μ A	80.28 μ A	0.35
g_m	887.68 μ S	891.5 μ S	0.43
g_{ds}	4.82 μ S	4.836 μ S	0.33
r_o	207.6 k Ω	206.8 k Ω	0.385
R_D	11.25 k Ω	11.25 k Ω	0.000
W	43.67 μ m	43.67 μ m	0.000
L	2 μ m	2 μ m	0.000

Comments

1) DC OP comparison with Part 1 (chart-based design):

Since the design used a chart extracted directly from simulation with the same technology and the same L , the operating point from the DC simulation matches the chart values closely. The drain current is approximately $80\text{ }\mu\text{A}$, confirming that the width scaling by ratio and proportion was correct. The region code reads 2 (saturation), as required for a linear amplifier.

2) r_o vs. R_D – is ignoring r_o justified?

From the DC OP, $r_{oQ} = 1/g_{ds,Q}$. At $L = 2\text{ }\mu\text{m}$, r_o is typically much larger than $R_D = 11.25\text{ k}\Omega$, so the error introduced by ignoring r_o in the gain formula $|A_v| \approx g_m R_D$ is small (a few percent). However, if minimum L were used (200 nm), r_o would drop significantly, and the parallel combination $R_D \parallel r_o$ would be noticeably smaller than R_D alone – the assumption would *not* be justified.

2.1.2 Intrinsic Gain and Amplifier Gain

The **intrinsic gain** of the transistor is defined as:

$$A_{\text{int}} = -g_m r_o = \frac{g_m}{g_{ds}} \approx -184 \quad (10)$$

This is the maximum possible voltage gain achievable from a single transistor (when $R_D \rightarrow \infty$, i.e., current-source load).

The **amplifier gain** with a resistive load is:

$$|A_v| = g_m (R_D \parallel r_o) = g_m \frac{R_D r_o}{R_D + r_o} \approx 9.5 \quad (11)$$

Since $r_o \gg R_D$ in the long-channel case, we have $R_D \parallel r_o \approx R_D$ and:

$$|A_v| \approx g_m R_D \ll g_m r_o = A_{\text{int}} \quad (12)$$

The amplifier gain is therefore **much less** ($\ll$) than the intrinsic gain in this design, because the resistive load R_D is much smaller than r_o .

2.1.3 AC Analysis Results

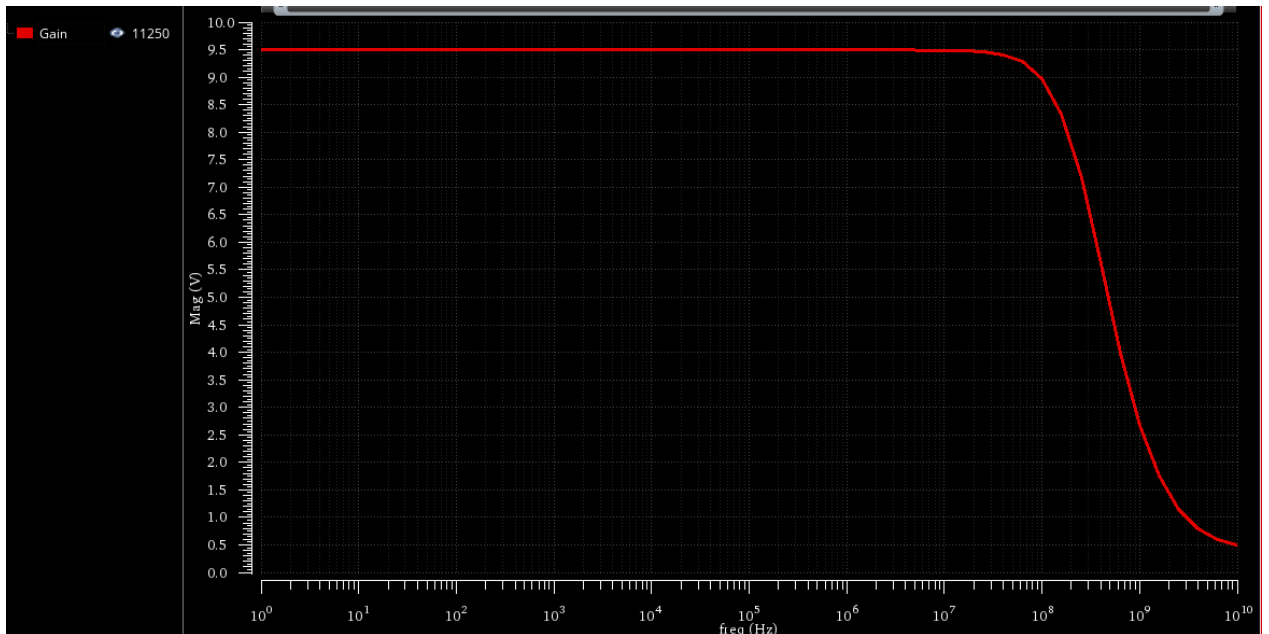


Figure 5: AC gain magnitude vs. frequency (1 Hz to 1 GHz).

IT1:LAB02_2:1	Gain		
IT1:LAB02_2:1	<code>ymin(mag(VF("/Vout")))</code>	9.512	

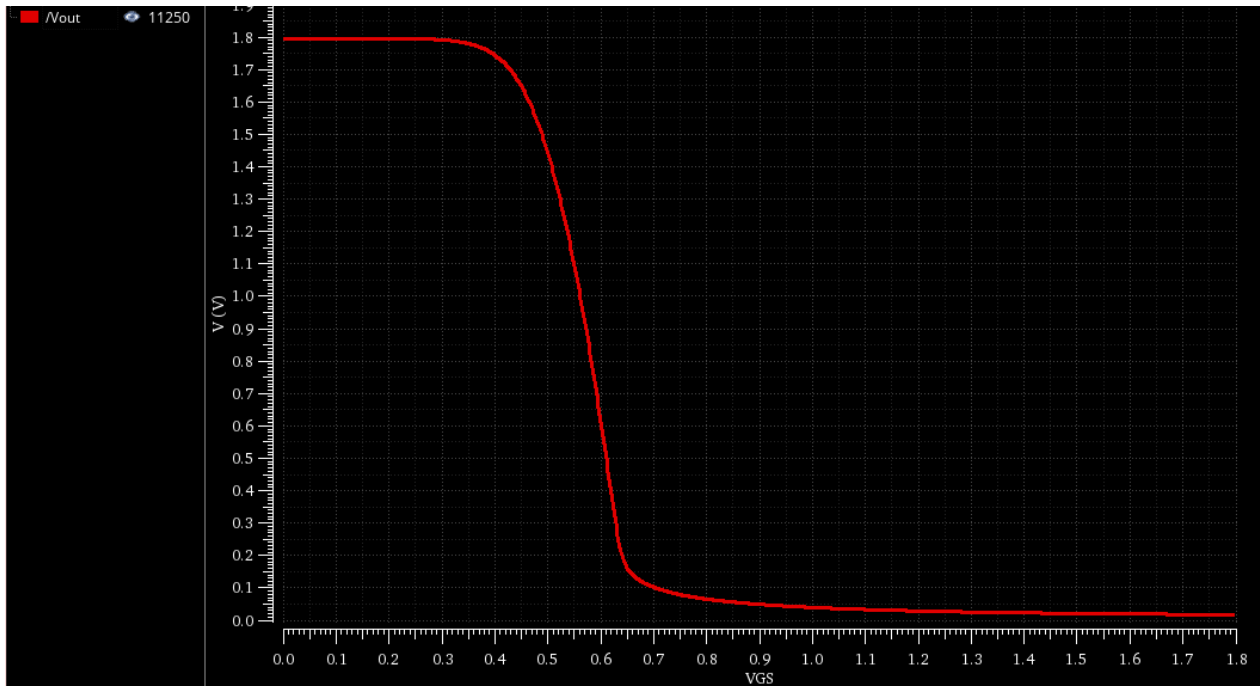
Figure 6: Annotated DC gain value from Cadence.

Comments

The simulated DC gain from AC analysis is annotated as ≈ 9.578 (approximately 10), which agrees with the specification within reasonable tolerances of the chart-based design methodology. The gain rolls off at high frequency due to the parasitic capacitances at the output node (C_{gd} , C_{db} , and any load capacitance).

2.2 Gain Non-Linearity

2.2.1 DC Sweep: V_{OUT} vs. V_{IN}

Figure 7: V_{OUT} vs. V_{IN} from DC sweep (0 to V_{DD} , 2 mV step).

Comments

Is the V_{OUT} - V_{IN} relation linear?

No. The transfer characteristic is not Linear. This is expected because:

- At low V_{IN} before V_{th} : the transistor is in cut-off; $V_{OUT} = V_{DD}$ (no current flows, no voltage drop across R_D).
- At the designed bias point $V_{in} < V_{GS} + V_{th}$: the transistor is in saturation; the curve has its maximum slope(linear), approximately equal to $-|A_v|$.
- At high V_{IN} , $V_{in} > V_{GS} + V_{th}$: the transistor enters triode, V_{DS} drops, and the gain collapses(non linear). V_{OUT} approaches 0.

The relationship is only approximately linear (quasi-linear) over a small range of V_{IN} around the bias point V_{GSQ} .

2.2.2 Derivative: Small-Signal Gain vs. V_{IN}

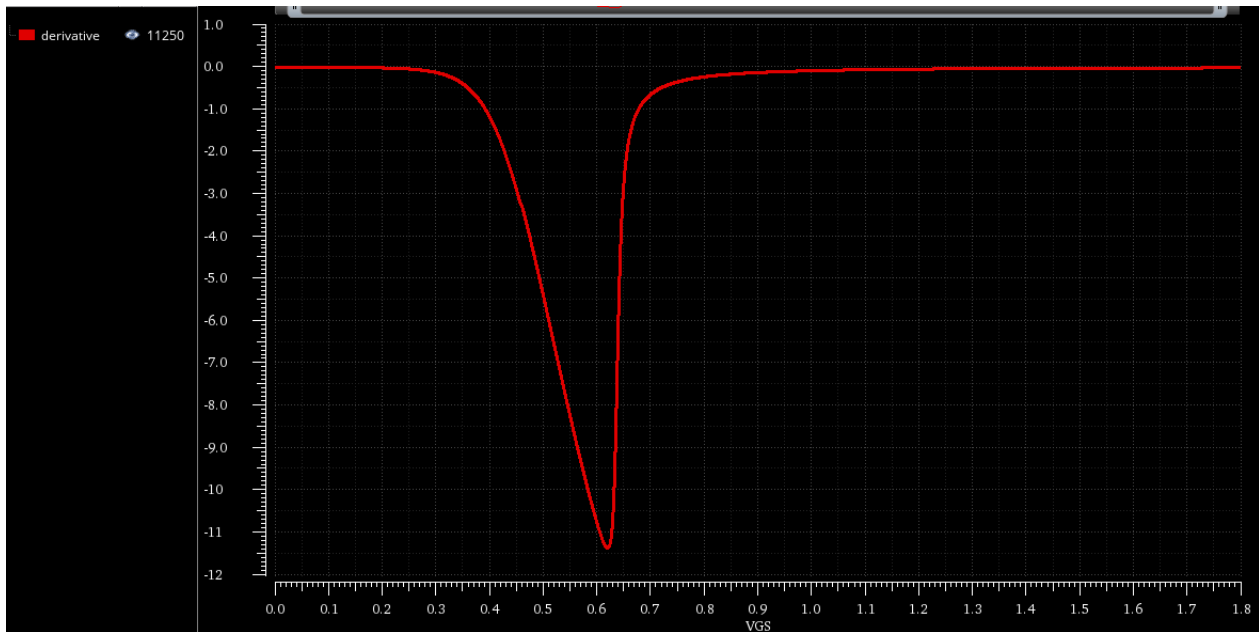


Figure 8: Derivative dV_{OUT}/dV_{IN} (instantaneous small-signal gain) vs. V_{IN} .

Comments

Is the gain linear (independent of input)?

No. The plot of dV_{OUT}/dV_{IN} vs. V_{IN} shows that the gain varies significantly with the input voltage. This is a direct consequence of the non-linear I_D - V_{GS} relationship of the MOSFET. Specifically:

- $g_m = \partial I_D / \partial V_{GS}$ is itself a function of V_{GS} (it is proportional to V_{ov} for the square-law device and saturates for short-channel devices).
- Since $\text{gain} = -g_m R_D$, any variation in g_m translates directly to gain variation.
- The gain is approximately constant only in a narrow range around the DC bias point.

2.2.3 Transient Simulation:

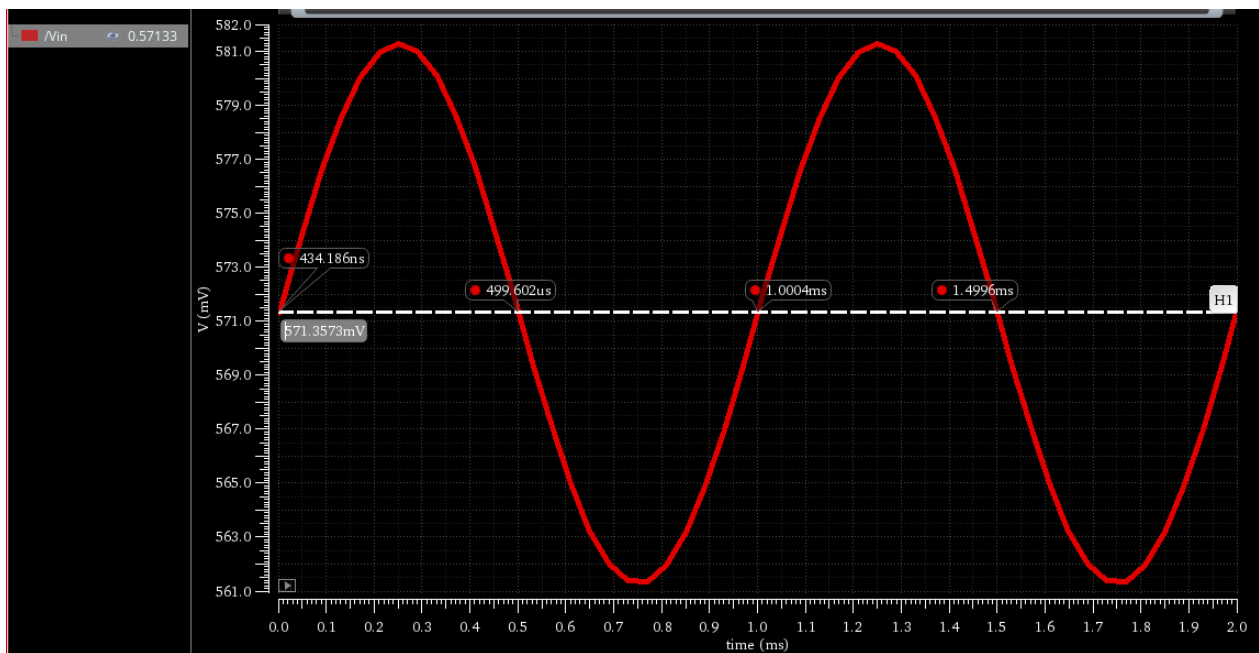


Figure 9: Input voltage V_{IN} vs. time.

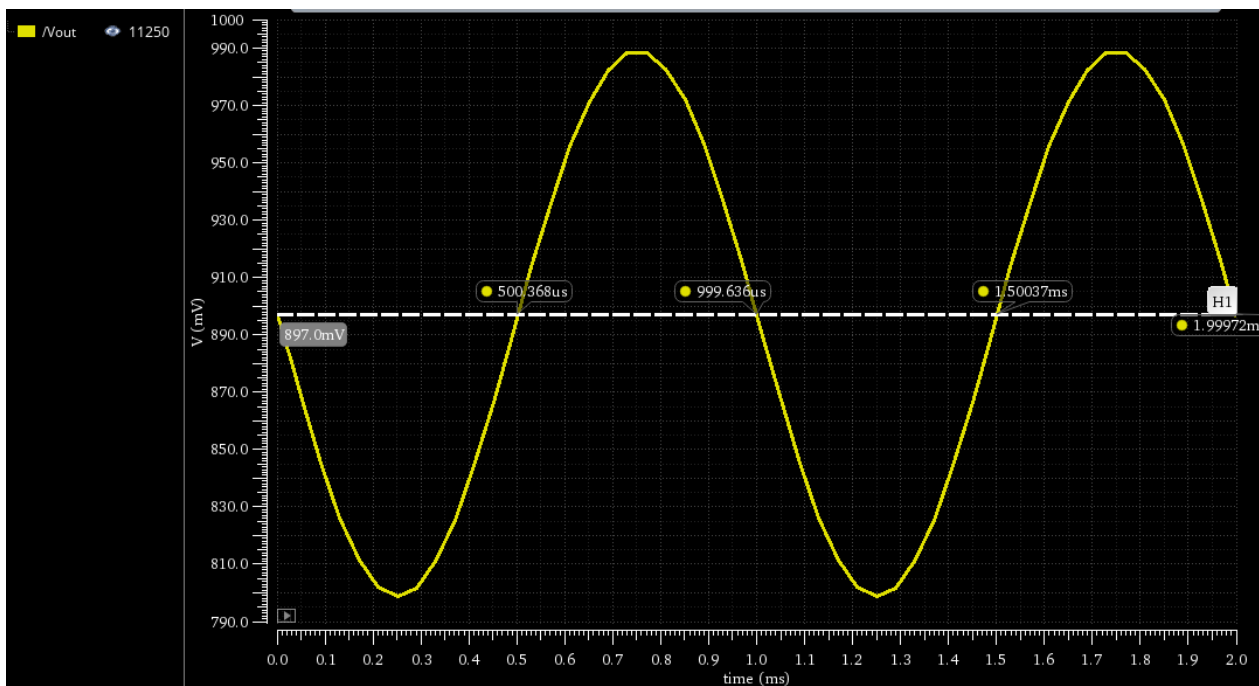


Figure 10: Output voltage V_{OUT} vs. time.

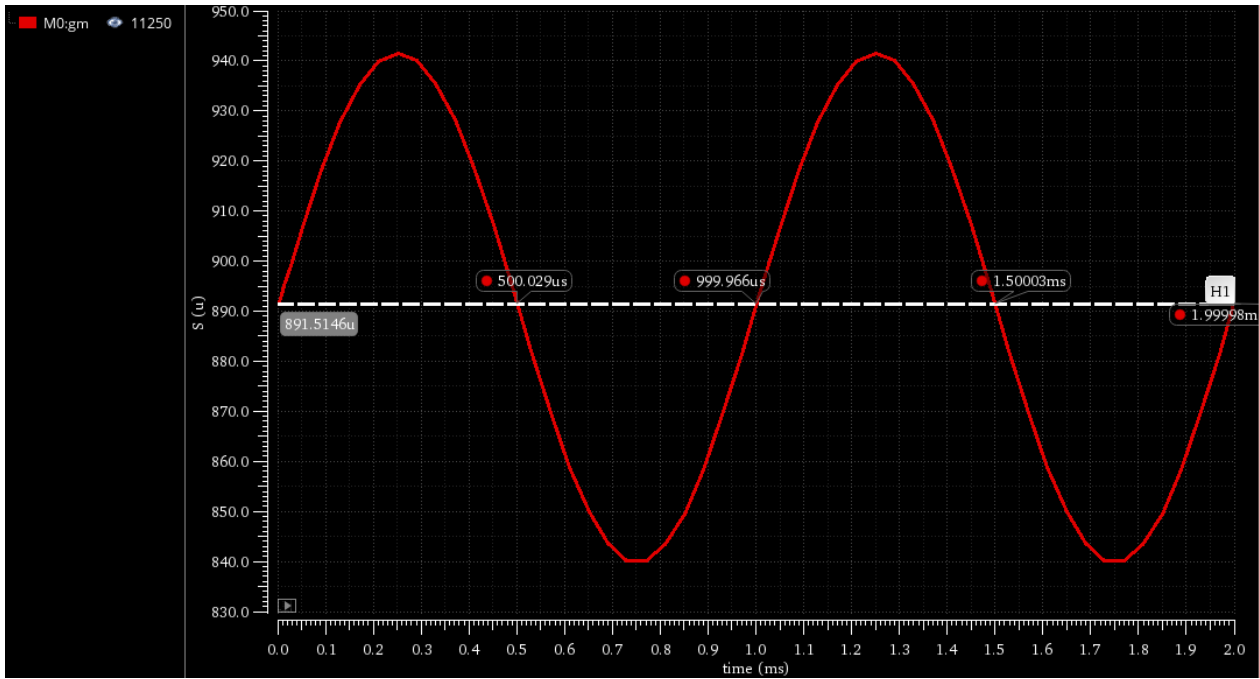


Figure 11: g_m vs. time under a 1 kHz, 10 mV sinusoidal input superimposed on V_{GSQ} .

Comments

Does g_m vary with the input signal?

Yes. The g_m vs. time plot shows that g_m oscillates sinusoidally around its DC bias value, in phase with the input sinusoid. This occurs because g_m depends on the instantaneous V_{GS} , which is the DC bias point plus the AC signal.

What does this mean? Is this amplifier linear?

A varying g_m means the instantaneous gain $A_v(t) = -g_m(t)R_D$ is time-varying. This introduces harmonic distortion: even a pure sinusoidal input produces an output that contains harmonics at $2f$, $3f$, etc. For a 10 mV amplitude the distortion is small (the amplifier is only mildly nonlinear), but it becomes severe for large signals. This amplifier is therefore **not linear** in the strict sense.

2.3 [Optional] Maximum Gain

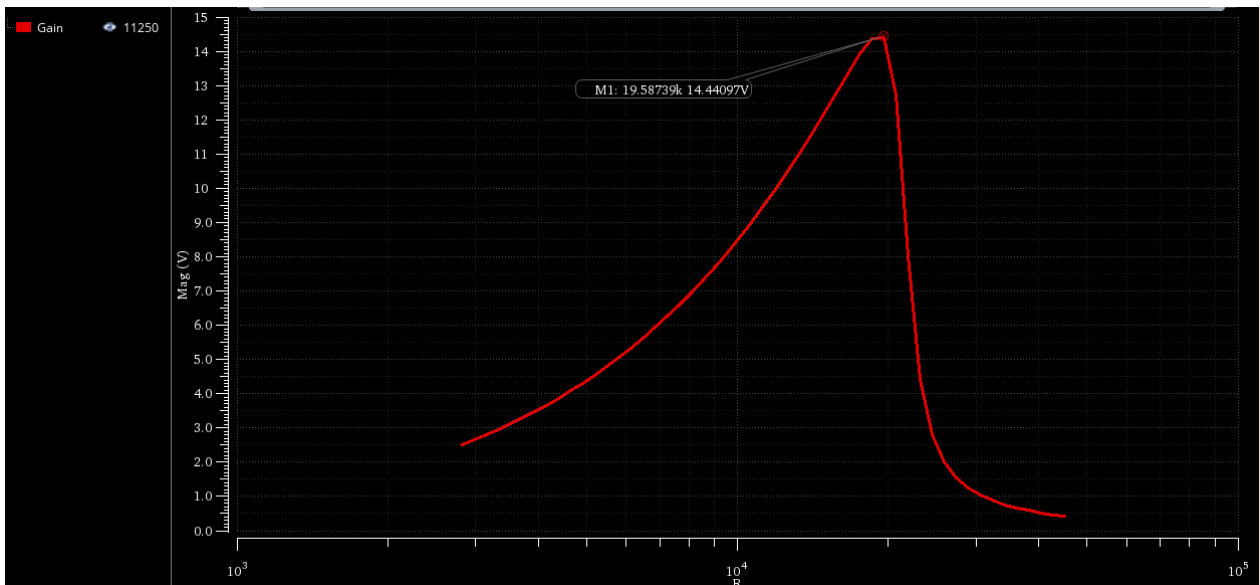


Figure 12: Gain vs RD

$$R_D^{(max)} = 19.587 K\Omega \quad (13)$$

and the corresponding maximum gain is:

$$|A_v|_{max} = 14.44 \quad (14)$$

2.3.1 Hand Analysis: Gain as a Function of R_D

The small-signal gain including r_o is:

$$|A_v(R_D)| = g_m(R_D || r_o) = g_m \frac{R_D r_o}{R_D + r_o} \quad (15)$$

Note, however, that **the bias current changes with R_D** when V_{GSQ} is held constant and the supply is fixed. With $V_{outCM} = V_{DD} - I_D R_D$, the bias point shifts as R_D varies; consequently g_m and r_o are themselves functions of R_D . This makes the analysis more involved. In the simplified case where g_m and r_o are treated as constants (fixed bias current):

- For $R_D \ll r_o$: $R_D || r_o \approx R_D$, so gain $\approx g_m R_D$ increases linearly with R_D .
- For $R_D \gg r_o$: $R_D || r_o \approx r_o$, so gain $\approx g_m r_o = A_{int}$, a constant.
- Maximum gain equals the intrinsic gain $A_{int} = g_m r_o$, achieved when $R_D \rightarrow \infty$.

In practice with a fixed V_{DD} and fixed V_{GSQ} , increasing R_D eventually pushes V_{DS} below V_{ov} (transistor leaves saturation), causing the gain to *drop*. So the gain peaks at an optimal R_D that keeps the transistor in saturation while maximising $g_m R_D$.

The condition for maximum gain under the constraint $V_{out,CM} = V_{DD} - I_D R_D \geq V_{ov}$ leads to the **maximum** R_D just before the transistor exits saturation:

$$R_D^{(max)} = \frac{V_{DD} - V_{ov}}{I_D} = 20 K\Omega \quad (16)$$

and the corresponding maximum gain is:

$$|A_v|_{max} = g_m R_D^{(max)} || r_o \approx g_m \frac{V_{DD} - V_{ov}}{I_D} = \frac{2(V_{DD} - V_{ov})}{V^*} = 18 \quad (17)$$

Parameter	Simulation	Hand Analysis	Error
Maximum Load Resistance ($R_D^{(max)}$)	19.587 K Ω	20 K Ω	2.065%
Maximum Gain ($ A_v _{max}$)	14.44	18	19.7%

Table 5: Comparison of Simulation and Hand Analysis Results

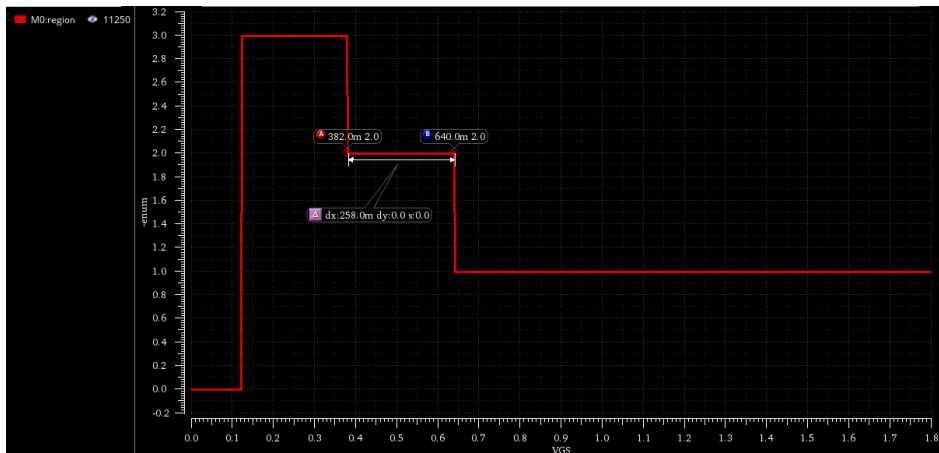


Figure 13: Region of NMOS vs VGS

the available swing equals **258 mV**.

Comments

Why does gain first increase then decrease with R_D ?

- **Increasing phase (R_D small):** $R_D \ll r_o$, so increasing R_D directly increases $g_m R_D$. The transistor stays in saturation.
- **Decreasing phase (R_D large):** As R_D grows, $V_{DS} = V_{DD} - I_D R_D$ decreases. Eventually $V_{DS} < V_{ov}$, pushing the transistor into triode, where g_m drops sharply and g_{ds} rises. The gain collapses.

Effect of supply scaling on gain:

Scaling down V_{DD} reduces the headroom available for the output swing. Since the maximum gain is $\propto (V_{DD} - V_{ov})$, a lower supply directly reduces the achievable gain.

2.4 [Optional] Gain Linearization via Feedback

2.4.1 PMOS Active Load Schematic

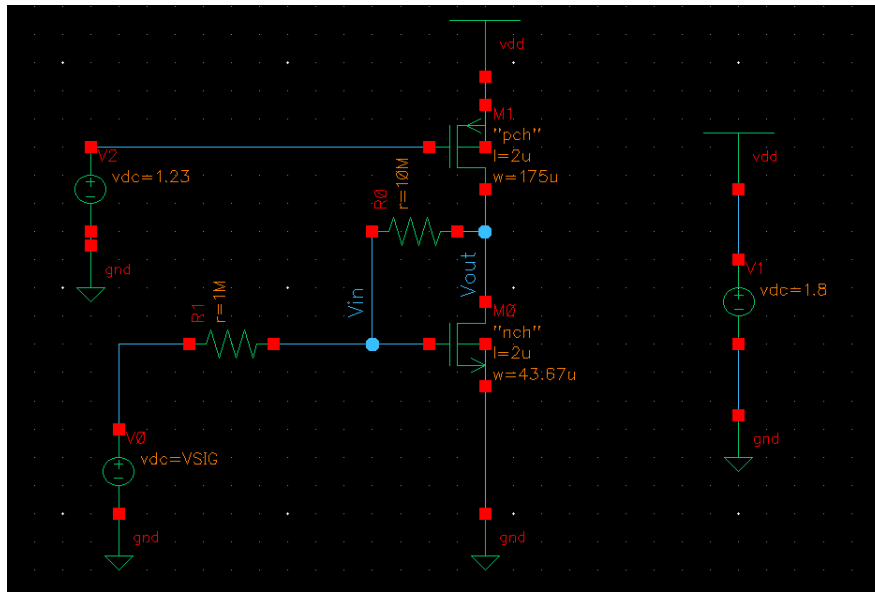
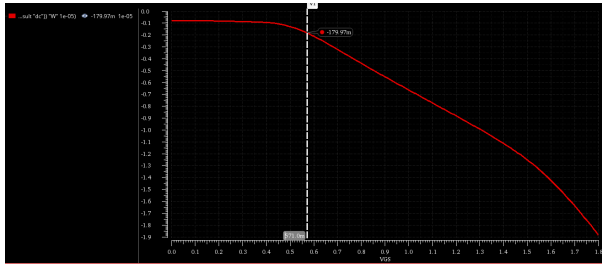
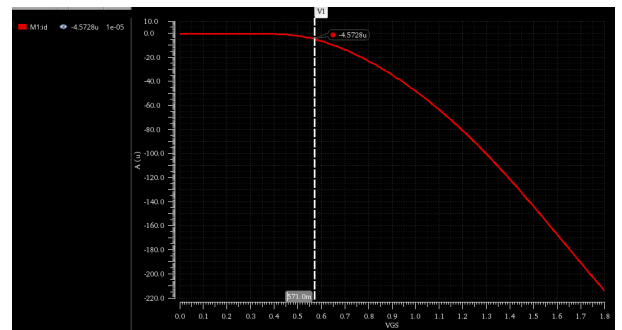


Figure 14: CS amplifier with PMOS current-source load and feedback resistors R_{in} and R_f for gain linearization.

2.4.2 PMOS Sizing Chart



(a) V^* vs. $|V_{GS}|$ for the reference PMOS ($W = 10 \mu\text{m}$, $L = 2 \mu\text{m}$).



(b) vs. $|V_{GS}|$ for the reference PMOS ($W = 10 \mu\text{m}$, $L = 2 \mu\text{m}$).

Figure 15: Characterization curves for the reference PMOS.

From the PMOS characterization charts, we extract the operating point values:

- $V_{GSX} = 571 \text{ mV}$
- $I_{DX} = 4.5728 \mu\text{A}$

2.4.3 PMOS Sizing

Using the PMOS sizing chart (Figure 15a), the same $V_Q^* = 0.18 \text{ V}$ as NMOS is assumed for the PMOS load. The PMOS must carry the same bias current $I_{DQ} = 80 \mu\text{A}$:

$$W_P = 10 \mu\text{m} \times \frac{I_{DQ}}{I_{DX,P}} \Big|_{V^*=0.18 \text{ V}} = 175 \mu\text{m} \quad (18)$$

2.4.4 Feedback Resistors

The feedback resistor R_f sets the closed-loop gain:

$$|A_{v,\text{closed}}| \approx \frac{R_f}{R_{in}} \implies R_f = |A_v| \times R_{in} = 10 \times 1 \text{ M}\Omega = 10 \text{ M}\Omega \quad (19)$$

With $R_{in} = 1 \text{ M}\Omega$, choose $R_f = 10 \text{ M}\Omega$.

2.4.5 DC Sweep with Feedback

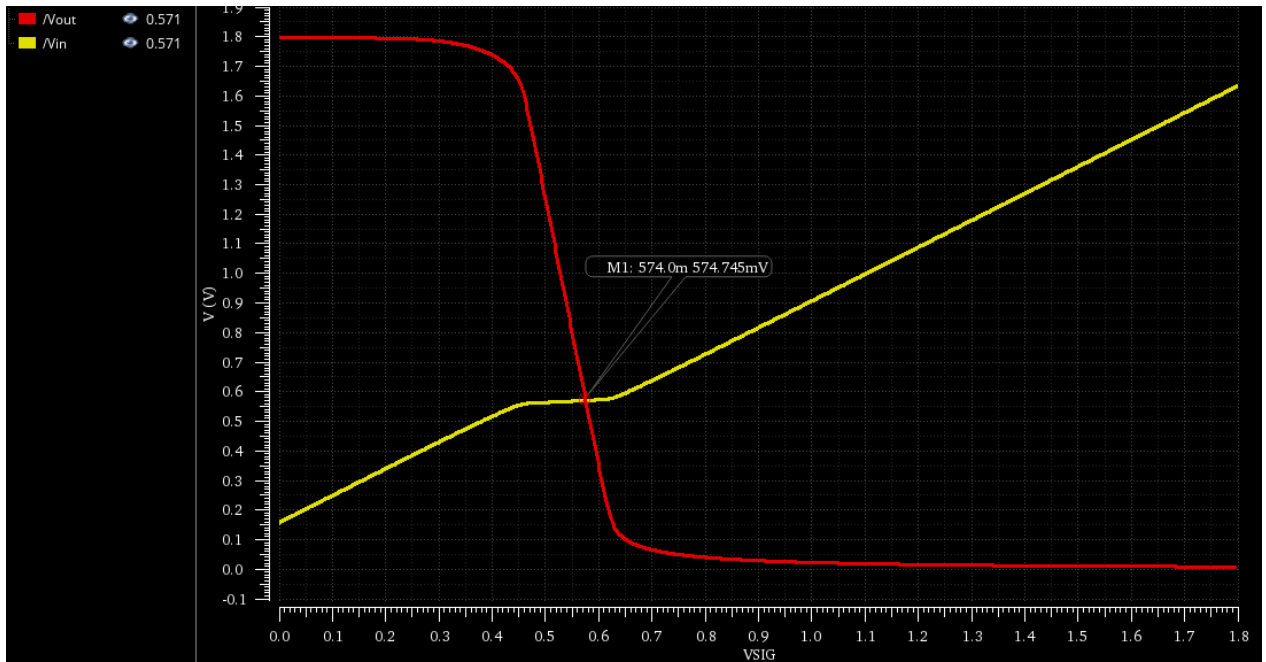


Figure 16: V_{IN} and V_{OUT} vs. V_{SIG} (DC sweep with PMOS load and feedback).

Comments

At what voltage do the V_{IN} and V_{OUT} curves cross?

They cross at $V_{SIG} = 574\text{mV}$ (the DC bias gate voltage of M1). $I_{DN} = I_{DP}$.

Is V_{OUT} vs. V_{SIG} more linear than before?

We can divide the relationship between V_{out} and V_{SIG} into three distinct operating regions:

- **Cut-off Mode:** When $V_{SIG} < V_{TH}$, the MOSFET is in cut-off, resulting in $V_{out} = V_{DD} \approx 1.8\text{ V}$.
- **Saturation Region:** When $V_{SIG} \geq V_{TH}$ and $V_{GD} \leq V_{TH}$, the MOSFET operates in the saturation region. Because feedback improves linearity, this can be approximated as a linear region (characterized by a high slope and high gain). As current flows through the transistor, V_{out} decreases due to the voltage drop across the active load.
- **Triode Region:** When $V_{SIG} > V_{TH}$ and $V_{GD} > V_{TH}$, the MOSFET enters the triode region. The gain drops significantly, resulting in a very small slope and minimal variation of V_{out} with respect to V_{SIG} .

Consequently, the relationship between V_{out} and V_{SIG} is not strictly linear across the entire curve.

2.4.6 Derivative (Gain) with Feedback

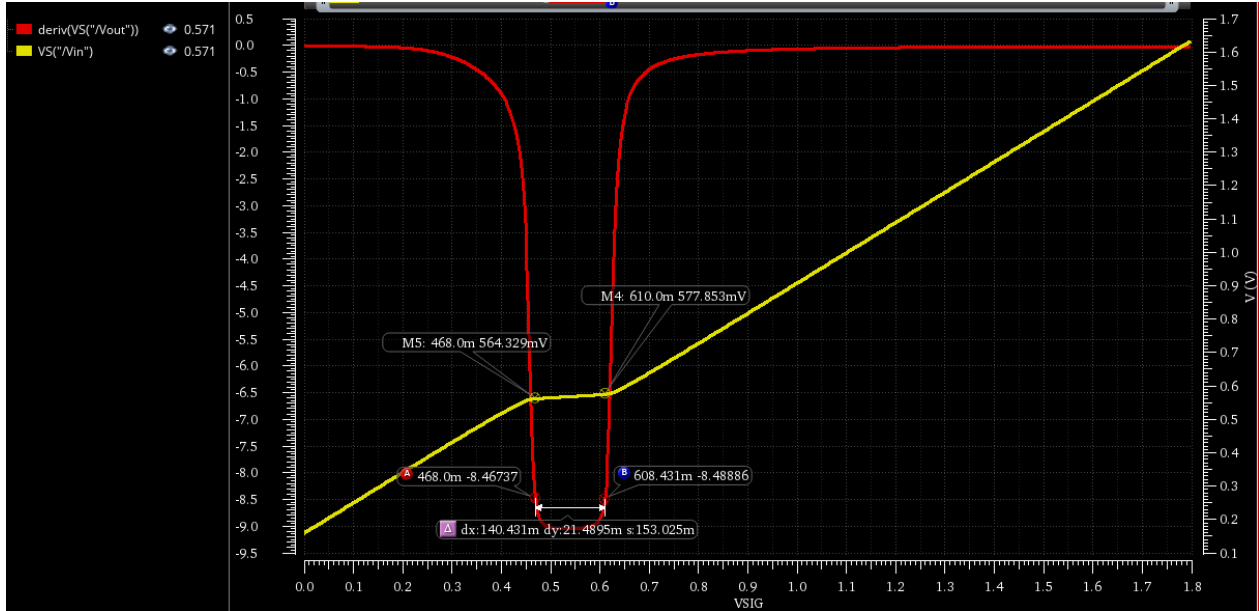


Figure 17: dV_{OUT}/dV_{SIG} (small-signal gain) vs. V_{SIG} with feedback.

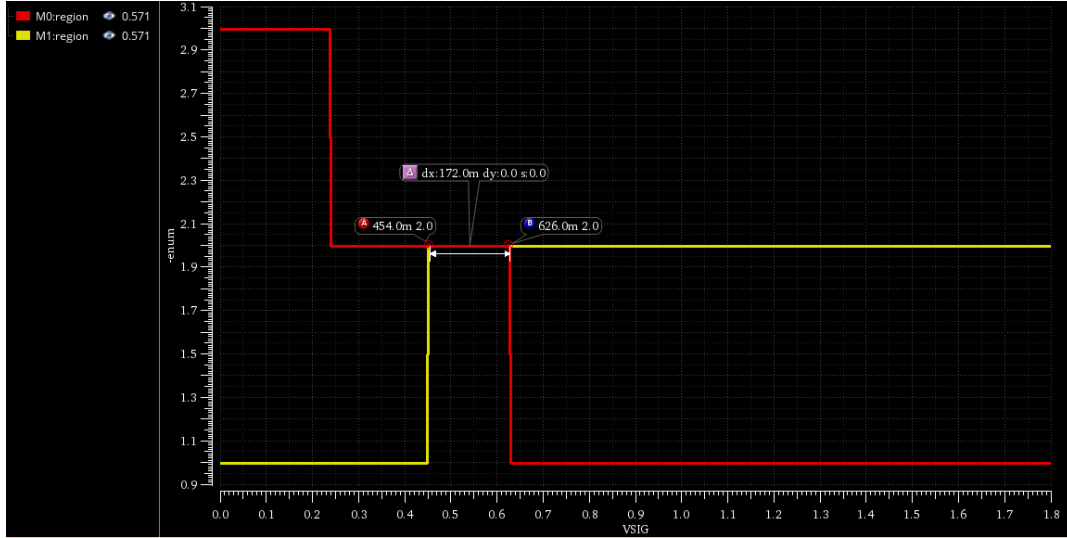


Figure 18: Regions of operations for NMOS and PMOS

Comments

Is the gain more constant with feedback?

Yes. The derivative plot shows a much flatter gain characteristic compared to the open-loop case. Feedback reduces gain non-linearity at the cost of reduced gain magnitude. The linear input range is extended because the feedback desensitizes the closed-loop gain to variations in the open-loop gain A_{OL} : as long as $A_{OL} \gg R_f/R_{in}$, the closed-loop gain $\approx R_f/R_{in}$ regardless of g_m fluctuations.

Analytical linear input range:

The amplifier fails when M1 or M2 exits saturation ($V_{DS} < V^*$). The output range before clipping is approximately $V_{DD} - 2V^* = 1.8 - 2(0.18) = 1.44$ V. The corresponding input range is:

$$\Delta V_{in} \approx \frac{V_{DD} - 2V^*}{|A_v|} = \frac{1.44}{10} = 144 \text{ mV} \quad (20)$$

The range for V_{SIG} where linear gain is from 510 mV to 634 mV is 0.134 V.

The range for V_{IN} where the gain is linear is from 594.5 mV to 606.4 mV is 11.9 mV and average value for V_{in} is 618 mV (it's the region where the derivative of V_{out} is constant). For NMOS to be in sat:

$$V_{GS} \geq 382.6 \text{ mV} \quad \text{and} \quad V_{out} \geq V_{ov} = V_Q^* = 180 \text{ mV}$$

$$\mathbf{V_{out} \geq 0.18 \text{ V}}$$

For PMOS to be in sat:

$$V_{SD} \geq V_{SG} - |V_{THP}|$$

$$V_{out} \leq V_G + |V_{THP}|$$

$$V_{out} \leq 1.2 + 0.393$$

$$\mathbf{V_{out} \leq 1.593 \text{ V}}$$

So the range of V_{out} for the linear gain is 1.413 V, then the range of the input that will be given by

$$\text{Range } V_{SIG} = \frac{\text{Range } V_{OUT}}{\text{Gain}} = \frac{1.413}{10} = 0.1413 \text{ V}$$

Using rule,

$$\text{Range } V_{SIG} = \frac{V_{DD} - 2V_Q^*}{\text{Gain}} = \frac{1.8 - 2 \times 0.18}{10} = 0.144 \text{ V}$$

$$\text{Error}\% = \frac{0.144 - 0.134}{0.144} \times 100 = 6.944\%$$